

STAT 318-01 Regression Analysis and Statistical Models

Spring 2026

Instructor: Prof. Qing (Wendy) Wang

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Lecture: Mondays & Thursdays at 9:55–11:10am and Wednesdays at 9:30am–10:20am

Classroom: Science Center L-039

In-person lectures start on Wednesday, January 21. There are three lectures every week, unless otherwise notified.

Attached Tutor: Nicky Sun (ys108)

Office Hours & Tutoring Sessions:

- Monday (tutor): 7:00-8:00 pm (in-person, SCI H-403)
- Wednesday: 1:30-2:30 pm (in-person, SCI H-401)
- Thursday (tutor): 5:15-6:15 pm (in-person, SCI-H403)
- Friday: 2:30–3:30 pm (virtual, [click here for the Zoom link](#))

Office hours start on Friday, January 23; tutoring sessions start on Friday, January 30.

Nicky Sun (ys108) is the attached tutor of STAT 318 this semester. In addition to her weekly tutoring sessions, she is also available for one-on-one tutoring, maximum 30 minutes per week. If you need additional help and would like to schedule an individual meeting with her, you could reach out to her by email (ys108) to schedule with her directly.

Important Dates:

January 21 (Wednesday): first class meeting

February 17 (Tuesday): Monday Schedule

March 16-20: no classes (Spring Break)

April 15 (Wednesday): no classes (Ruhlman Conference)

April 20 (Monday): no classes (Patriot's Day)

April 27, April 29, and April 30: in-class presentations

Almost all problem sets are due at 11:59pm on Fridays, unless otherwise noted. To submit your completed assignment online, you need to upload *a single PDF document as an attachment* through the Assignment tab on the course website.

Prerequisite: MATH 205, MATH 206, and STAT 218 (or STAT 260 or an intro stats course).

This course is not suited for students who have not taken linear algebra or introductory statistics. *Prior experience with programming in the R environment is expected.*

Recommended Textbooks:

- *A Second Course in Regression Analysis* (7th edition), by Mendenhall and Sincich. Publisher: Pearson.

The lectures mostly follow the content of this textbook. However, this book does not include linear-algebra-based derivations that are required for this course. Note that the class handouts

and course materials posted on the course website include all the details you need to know that complement the content of this book.

Two physical copies of the textbook is available on course reserve in the Clapp Library.

- *Applied Linear Regression Models* (5th edition), by Kutner, Nachtsheim, and Neter. Publisher: McGraw-Hill Irwin.

This book contains most mathematical derivations required for this course.

Course Description:

STAT 318 is an applied linear regression course that involves “hands on” data analysis. Topics covered during the semester include multiple linear regression models and their matrix representations, model diagnostics and remedial measures, model comparison and selection, and logistic regression and generalized linear regression models. A tentative course coverage can be found at the end of the syllabus (the chapter numbers follow the book by Mendenhall and Sincich). This course counts as a 300-level elective for the math major, math minor, statistics minor, and data science major.

Distribution: Mathematical Modeling and Problem Solving

Learning Goals: After taking STAT 318, you should be able to

- * choose and fit an appropriate regression model to study the association between variables given a real data set;
- * diagnose regression assumptions and employ remedial measures if some assumptions are violated;
- * compare the predictive performance of different statistical models;
- * identify the best subset of predictor variables using variable selection techniques;
- * understand the mathematical properties behind statistical methods;
- * program using statistical language R to realize model fitting, simulation, and graphical display;
- * learn about the importance and different aspects of statistics/data ethics.

Use R!

R is an open source statistical programming language. The R language is the most used language among statisticians for data analysis, statistical simulations, and graphical display. R will be used throughout the semester for realizing linear (or generalized linear) regressions, model diagnostics, model comparison and selection, etc.

R can be downloaded from <http://www.cran.r-project.org/>. For a more detailed introduction of R, please refer to the document <http://www.cran.r-project.org/doc/manuals/R-intro.pdf>. You may also consider downloading R Studio which provides an integrated development environment (IDE) and a user-friendly interface.

Some useful R/R Studio resources can be found on the course website in the folder “R Help”.

Grading:

Category	Percentage
Homework	20%
Midterm (2)	40%
Case Study (2)	20%
Final Project	10%
Final Presentation	10%
Total	100%

- **Homework:**

The homework is worth 20% of the semester grade and consists of 8 assignments. All homework assignments, except the last one, are due on Fridays at 11:59pm and must be submitted online via the *Assignment tab* on the course website. Each problem set is graded on a 100-point scale. No late submission is accepted unless the student has permission from the instructor at least 24 hours in advance of the due date.

Late Submission Policy:

- In general, late submission is not accepted unless a student has requested an extension at least 24 hours in advance.
- There is a one-time “forgiveness” policy. Specifically, if a student accidentally misses a deadline by 30 minutes or less, their homework is accepted with a late penalty of 5 points off; submitting a homework 30 minutes to 1 hour late will incur 10 points deduction. No problem set is accepted 1 hour after the deadline, unless an extension has been granted in advance.

You are encouraged to discuss homework questions with others, but you must submit your own work. Whenever R is used to solve some homework questions, you may copy and paste your R commands and output into a word document and include necessary explanations or comments. *Another convenient way to type up your answers to homework questions and include R command/output together is by using R MARKDOWN. A simple template and instruction on how to use R MARKDOWN can be found on the course website. Using R MARKDOWN for your problem sets is encouraged but not required.*

When submitting your completed assignment electronically online, it's preferred that you combine all the pages into a single PDF document.

- **Exams:**

There are two in-class midterm exams; each counts for 20% of the semester grade. You are allowed to bring one-page, *single-sided* of notes and a simple calculator to the in-class exam. Besides these, no other aid or materials are allowed during the test. The first midterm (Topics 1-4) is tentatively scheduled on **March 12** (Thursday), and the second midterm (Topics 5-8) is on **April 16** (Thursday). A sample practice exam and its solution will be posted on the course website before each scheduled in-class midterm to help you better prepare for the test.

Conflicts on exam dates must be resolved in advance. An unexcused absence on an exam date may result in a score of 0 in the test. Please allow 24 hours for email response.

- **Case Studies:**

There will be two take-home case studies during the semester. The case study is a real data analysis assignment and each counts for 10% of the semester grade. Students need to complete them *independently*. Each of them will be assigned after the in-class midterm exam. The first case study will be assigned after the completion of Topic 5 (tentatively due on March 24), and the second case study is tentatively due on April 21. The case study provides you with the opportunity to address a research question by analyzing a real data set using statistical modeling tools learned in class, and help hone your (written) communication skills in presenting statistical results.

- **Final Project and Final Presentation:**

There is no written final exam of this course. Instead, students will work in *groups of two* to complete a team project. The goal of the project is to introduce you the challenge and excitement of real data analysis. Each team will find your own data and propose one or more research questions of interest. You need to apply appropriate statistical modeling techniques learned in class to analyze the data and draw conclusions. More instructions and requirements on the final project and presentation can be found on the course website.

The project presentation in STAT 318 no longer satisfies the presentation requirement of the mathematics major.

- **Acceptable/Unacceptable use of AI:**

- Generative AI presents a powerful tool that can significantly enhance one's learning experience. There are numerous effective ways to leverage generative AI to facilitate learning, such as requesting examples of R functions, or seeking clarification on complex statistical concepts. However, it is crucial to recognize the limitations of AI and avoid over-reliance on it. Below are occasions where the use of AI is not acceptable in this course.
- You should never upload a dataset from our homework into generative AI, or direct generative AI to use a particular dataset from any source.
- You should not copy your R code into generative AI, for debugging or any other reason.
- You should not copy homework question into generative AI.
- It is not acceptable to submit either written text or R code that you've copied from the output of generative AI, even if edited.

- **Learning Beyond the Classroom**

Besides studying the materials of STAT 318, there are a couple other ways to learn statistics outside the classroom.

1. Attend the statistics colloquium in the spring semester. More details about the talk will be announced later.
2. Submit your project written report to participate in the Undergraduate Class Project Competition—this is required for your final project. You should select the category “an intermediate course in statistics or data science”. Additionally, you need to enter my name as your faculty sponsor. So, I will receive a list of all submitted projects under my

sponsorship.

“The winning projects will be featured on the CAUSE web site and announced in the Amstat News journal published by the American Statistical Association. The authors of the winning projects may also be invited to present their work at the electronic undergraduate statistics conference. Cash prizes will be awarded to first, second and third place.” (<https://www.causeweb.org/usproc/>)

Disability accommodations: Any students who need disability accommodations may reach out to the Office of Accessibility and Disability Resources, preferably during the first 2 weeks of the semester, so that your request can be reviewed and approved in a timely manner. More information is available at the ADR’s website <https://www.wellesley.edu/adr>.

Announcement:

Syllabus, lecture notes, homework solutions, and all course materials will be posted on the course website (Sakai). You are expected to check this site frequently.

Topics and Course Coverage

	Topics	Reading (Mendenhall and Sincich, 7th edition)
1	Review of basic concepts	Chapter 1
2	Simple linear regression <i>Activity: verify unbiasedness of LS solution by simulation</i>	Chapter 2 & Chapter 3.1-3.5
3	Inference for simple linear regression Maximum likelihood method*	Chapter 3.6-3.11
4	Multiple linear regression Matrix representation*	Chapter 4.1-4.9
5	More about multiple linear regression <i>Activity: partial F-test exercises using R</i>	Chapter 4.10-4.14 & Chapter 5.1-5.5 5.7-5.10
6	Variable screening methods Leave-one-out and K -fold cross-validation* <i>Activity: implementation of cross-validation techniques</i>	Chapter 6
7	Regression pitfalls Principal component analysis* Power transformation (Box-Cox transformation)*	Chapter 7.1-7.6
8	Diagnostics of regression assumptions	Chapter 8.1-8.6
9	Logistic regression Generalized linear regression*	Chapter 9.5-9.6

* marked are topics outside the textbook (Mendenhall and Sincich) coverage but will be introduced in class.